

# Asset Cash Intensity and the Cross Section of Stock Returns

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October 21, 2025

## Abstract

This paper studies the asset pricing implications of Asset Cash Intensity (ACI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on ACI achieves an annualized gross (net) Sharpe ratio of 0.48 (0.42), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 36 (31) bps/month with a t-statistic of 3.59 (3.13), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in equity to assets, Growth in book equity, Change in current operating liabilities, Abnormal Accruals, Operating Cash flows to price, Asset growth) is 19 bps/month with a t-statistic of 2.00.

# 1 Introduction

Financial markets process vast amounts of information daily, yet mounting evidence suggests that prices do not immediately and fully reflect all available information. This deviation from the strong-form efficient market hypothesis creates predictable patterns in asset returns that persist across time and markets. Despite decades of research documenting hundreds of return predictors, fundamental questions remain about which firm characteristics generate robust return predictability and through what economic mechanisms these patterns arise.

Asset Cash Intensity (ACI), defined as the ratio of cash and cash equivalents to total assets, represents a fundamental aspect of firm financial policy that may contain valuable information about future returns. While the finance literature extensively examines cash holdings in the context of corporate finance decisions [Opler et al. \(1999\)](#); [Bates et al. \(2009\)](#), the asset pricing implications of cash intensity remain underexplored. We investigate whether ACI predicts cross-sectional stock returns and test whether this predictability arises from the gradual diffusion of information across market participants.

The slow diffusion of information provides a compelling framework for understanding why ACI might predict returns. When investors have limited attention, they cannot process all available information simultaneously, leading to systematic underreaction to value-relevant signals [Hirshleifer et al. \(2009\)](#); [DellaVigna and Pollet \(2009\)](#). This gradual information incorporation creates return continuation as prices slowly converge to fundamental values. The seminal work of [Hong and Stein \(1999\)](#) demonstrates that information diffuses gradually across investors, with momentum profits concentrated among stocks with low analyst coverage where information constraints bind most tightly.

ACI contains information about firms' operational efficiency, financial flexibility, and investment opportunities that investors may not immediately recognize. High

cash intensity signals conservative financial management and potential agency costs, while low cash intensity may indicate aggressive investment policies or financial constraints [Jensen \(1986\)](#); [Myers and Rajan \(1998\)](#). The complexity of interpreting cash holdings in different economic contexts creates information processing frictions that delay price adjustment. Investors must evaluate whether high cash represents precautionary savings, lack of investment opportunities, or managerial entrenchment, requiring careful analysis that extends beyond simple financial ratios.

The gradual diffusion hypothesis predicts that ACI's return predictability should concentrate where attention constraints and information frictions are most severe. Small firms with limited analyst coverage experience slower information diffusion, as documented by [Hong et al. \(2000\)](#) and [Zhang \(2006\)](#). These stocks receive less media attention and institutional investor scrutiny, allowing information in ACI to be incorporated more slowly into prices. Furthermore, retail investors, who dominate trading in smaller stocks, exhibit stronger behavioral biases and slower information processing [Barber and Odean \(2008\)](#). This framework generates testable predictions: ACI should predict returns more strongly among small, neglected stocks where information diffuses slowly, and the return predictability should manifest as continuation rather than reversal.

Our empirical analysis reveals that ACI strongly predicts cross-sectional stock returns. A value-weighted long/short trading strategy based on ACI achieves an annualized gross Sharpe ratio of 0.48, placing it in the top 13% of documented anomalies. The strategy generates a monthly average abnormal gross return relative to the Fama-French six-factor model of 36 basis points with a t-statistic of 3.59, demonstrating robust performance that cannot be explained by exposure to standard risk factors. The economic magnitude remains substantial after accounting for transaction costs, with the net Sharpe ratio of 0.42 exceeding 97% of anomalies in the factor zoo.

The predictive power of ACI exhibits patterns consistent with gradual information diffusion. Among the largest stocks—those with market capitalization above the 80th NYSE percentile where information typically flows most efficiently—the ACI strategy still achieves an average return of 32 basis points per month with a t-statistic of 2.12. This finding suggests that even in the most liquid segment of the market, investors underreact to information contained in cash intensity. The strategy’s alpha relative to the six-factor model among large-cap stocks equals 37 basis points monthly, indicating that the return predictability cannot be attributed to loadings on size, value, profitability, investment, or momentum factors.

Controlling for closely related signals strengthens our interpretation that ACI captures unique information about future returns. When we simultaneously control for six closely related anomalies—including operating cash flows to price, asset growth, and abnormal accruals—plus the six Fama-French factors, the ACI trading strategy maintains an alpha of 19 basis points per month with a t-statistic of 2.00. This result demonstrates that ACI contains incremental information beyond existing predictors of returns. The strategy’s consistent performance across different portfolio construction methodologies, with twenty out of twenty-five specifications producing t-statistics exceeding two, further confirms the robustness of our findings.

This paper makes several contributions to the asset pricing literature. First, we document a new cross-sectional return predictor that exhibits economically and statistically significant predictive power comparable to established anomalies. While [Palazzo \(2012\)](#) examine cash holdings in the context of aggregate market returns and [Simutin \(2013\)](#) study cash holdings’ role in financial constraints, our paper is the first to systematically investigate asset cash intensity as a cross-sectional return predictor through the lens of information diffusion. Our findings extend the literature on firm fundamentals and stock returns documented in the *Journal of Financial Economics* by [Cooper et al. \(2008\)](#) and [Novy-Marx \(2013\)](#).

Second, we provide new evidence supporting the gradual information diffusion hypothesis as a mechanism generating return predictability. Our results complement the findings in the Review of Financial Studies by [Hong and Stein \(2007\)](#) on slow information diffusion and extend the work in the Journal of Finance by [Cohen and Frazzini \(2008\)](#) on economic links and predictable returns. The persistence of ACI’s predictive power among large-cap stocks challenges pure risk-based explanations and suggests that even sophisticated investors underreact to information embedded in cash intensity. This evidence aligns with recent work in the Journal of Accounting Research by [Lee et al. \(2019\)](#) showing that fundamental signals predict returns through investor underreaction rather than risk.

Third, our comprehensive analysis using the [Novy-Marx and Velikov \(2023b\)](#) protocol establishes ACI as one of the most robust anomalies in the factor zoo. The strategy’s performance after transaction costs, its incremental predictive power beyond related anomalies, and its ability to enhance combination strategies demonstrate practical relevance for investors. These findings have important implications for understanding market efficiency and the limits of arbitrage. The persistence of return predictability associated with a simple, observable characteristic like cash intensity suggests that informational frictions remain substantial even in modern markets with sophisticated investors and advanced information technology.

## 2 Data

We obtain financial statement data from COMPUSTAT, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations using fiscal year-end values to ensure consistency in measurement across firms with different reporting

periods. signal construction relies on two key accounting variables. Operating activities net cash flow (OANCF) represents the cash generated or consumed by a firm’s core business operations during the fiscal year, calculated using the direct or indirect method as reported in the statement of cash flows. This measure captures the actual cash inflows and outflows from operating activities, excluding investing and financing activities. Property, plant, and equipment gross (PPEGT) represents the total historical cost of a firm’s tangible fixed assets before accumulated depreciation, including land, buildings, machinery, equipment, and other physical assets used in operations. construct the Asset Cash Intensity signal by computing the year-over-year change in operating cash flow scaled by lagged gross property, plant, and equipment:  $(\text{OANCF}(t) - \text{OANCF}(t-1)) / \text{PPEGT}(t-1)$ . This ratio captures the change in cash generation efficiency relative to the firm’s tangible asset base, measuring how effectively a firm converts its physical capital into operating cash flows over time. We require non-missing values for OANCF in consecutive years and PPEGT in the lagged period to compute the signal. To mitigate the influence of outliers, we exclude observations where PPEGT equals zero or is negative, as these cases would result in undefined or economically meaningless ratios. The resulting measure provides an annual indicator of changes in asset productivity from a cash flow perspective, with higher values suggesting improving cash generation relative to the firm’s investment in physical assets.

### 3 Signal diagnostics

Figure 1 plots descriptive statistics for the ACI signal. Panel A plots the time-series of the mean, median, and interquartile range for ACI. On average, the cross-sectional mean (median) ACI is -0.38 (0.01) over the 1989 to 2024 sample, where the starting date is determined by the availability of the input ACI data. The signal’s

interquartile range spans -0.33 to 0.30. Panel B of Figure 1 plots the time-series of the coverage of the ACI signal for the CRSP universe. On average, the ACI signal is available for 6.20% of CRSP names, which on average make up 7.29% of total market capitalization.

## 4 Does ACI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on ACI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high ACI portfolio and sells the low ACI portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short ACI strategy earns an average return of 0.33% per month with a t-statistic of 2.84. The annualized Sharpe ratio of the strategy is 0.48. The alphas range from 0.29% to 0.40% per month and have t-statistics exceeding 2.47 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.56, with a t-statistic of -8.82 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 432 stocks and an average market capitalization of at least \$2,146 million.

Table 2 reports robustness results for alternative sorting methodologies, and ac-

counting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 14 bps/month with a t-statistics of 1.97. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty exceed two, and for five exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -11-30bps/month. The lowest return, (-11 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -1.23. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the ACI trading strategy improves the achievable mean-variance efficient frontier



spanned by the factor models in twenty cases, and significantly expands the achievable frontier in thirteen cases.

Table 3 provides direct tests for the role size plays in the ACI strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and ACI, as well as average returns and alphas for long/short trading ACI strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80<sup>th</sup> NYSE percentile), the ACI strategy achieves an average return of 32 bps/month with a t-statistic of 2.12. Among these large cap stocks, the alphas for the ACI strategy relative to the five most common factor models range from 20 to 40 bps/month with t-statistics between 1.34 and 3.09.

## 5 How does ACI perform relative to the zoo?

Figure 2 puts the performance of ACI in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.<sup>1</sup> The vertical red line shows where the Sharpe ratio for the ACI strategy falls in the distribution. The ACI strategy’s gross (net) Sharpe ratio of 0.48 (0.42) is greater than 87% (97%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the ACI strategy (red line).<sup>2</sup> Ignoring trading costs, a \$1 invested in the ACI strategy

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<sup>1</sup>The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

<sup>2</sup>The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that

would have yielded \$2.43 which ranks the ACI strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the ACI strategy would have yielded \$1.94 which ranks the ACI strategy in the top 2% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the ACI relative to those. Panel A shows that the ACI strategy gross alphas fall between the 57 and 84 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 198906 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The ACI strategy has a positive net generalized alpha for five out of the five factor models. In these cases ACI ranks between the 77 and 95 percentiles in terms of how much it could have expanded the achievable investment frontier.

## 6 Does ACI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations.

Figure 5 plots a name histogram of the correlations of ACI with 210 filtered anomaly

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a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

signals.<sup>3</sup> Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price ACI or at least to weaken the power ACI has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of ACI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on  $\beta_{ACI}$  from Fama-MacBeth regressions of the form  $r_{i,t} = \alpha + \beta_{ACI}ACI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$ , where  $X$  stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on  $\alpha$  from spanning tests of the form:  $r_{ACI,t} = \alpha + \beta r_{X,t} + \epsilon_t$ , where  $r_{X,t}$  stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on ACI. Stocks are finally grouped into five ACI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ACI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on ACI and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the  $R^2$  from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the ACI signal in these Fama-MacBeth regressions exceed

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<sup>3</sup>When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

0.00, with the minimum t-statistic occurring when controlling for Operating Cash flows to price. Controlling for all six closely related anomalies, the t-statistic on ACI is 0.17.

Similarly, Table 5 reports results from spanning tests that regress returns to the ACI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the ACI strategy earns alphas that range from 25-37bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.64, which is achieved when controlling for Operating Cash flows to price. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the ACI trading strategy achieves an alpha of 19bps/month with a t-statistic of 2.00.

## 7 Does ACI add relative to the whole zoo?

Finally, we can ask how much adding ACI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 158 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 158 anomalies augmented with the ACI signal.<sup>4</sup> We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 158-anomaly combination strategy grows to \$31.75, while \$1 investment in the combination strategy

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<sup>4</sup>We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which ACI is available.

that includes ACI grows to \$37.41.

## 8 Conclusion

This study provides robust evidence that Asset Cash Intensity (ACI) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous testing protocol of Novy-Marx and Velikov (2023), demonstrates that ACI-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on ACI delivers impressive performance metrics, with an annualized gross Sharpe ratio of 0.48 and net Sharpe ratio of 0.42 after accounting for transaction costs. The strategy’s resilience is particularly noteworthy, generating statistically significant abnormal returns of 36 basis points per month ( $t$ -statistic = 3.59) relative to the Fama-French five-factor model augmented with momentum. Even more compelling is the signal’s ability to maintain economic significance when confronted with six closely related strategies from the factor zoo, still producing a monthly alpha of 19 basis points ( $t$ -statistic = 2.00). This suggests that ACI captures unique information about firm fundamentals not fully explained by existing factors or similar accounting-based measures.

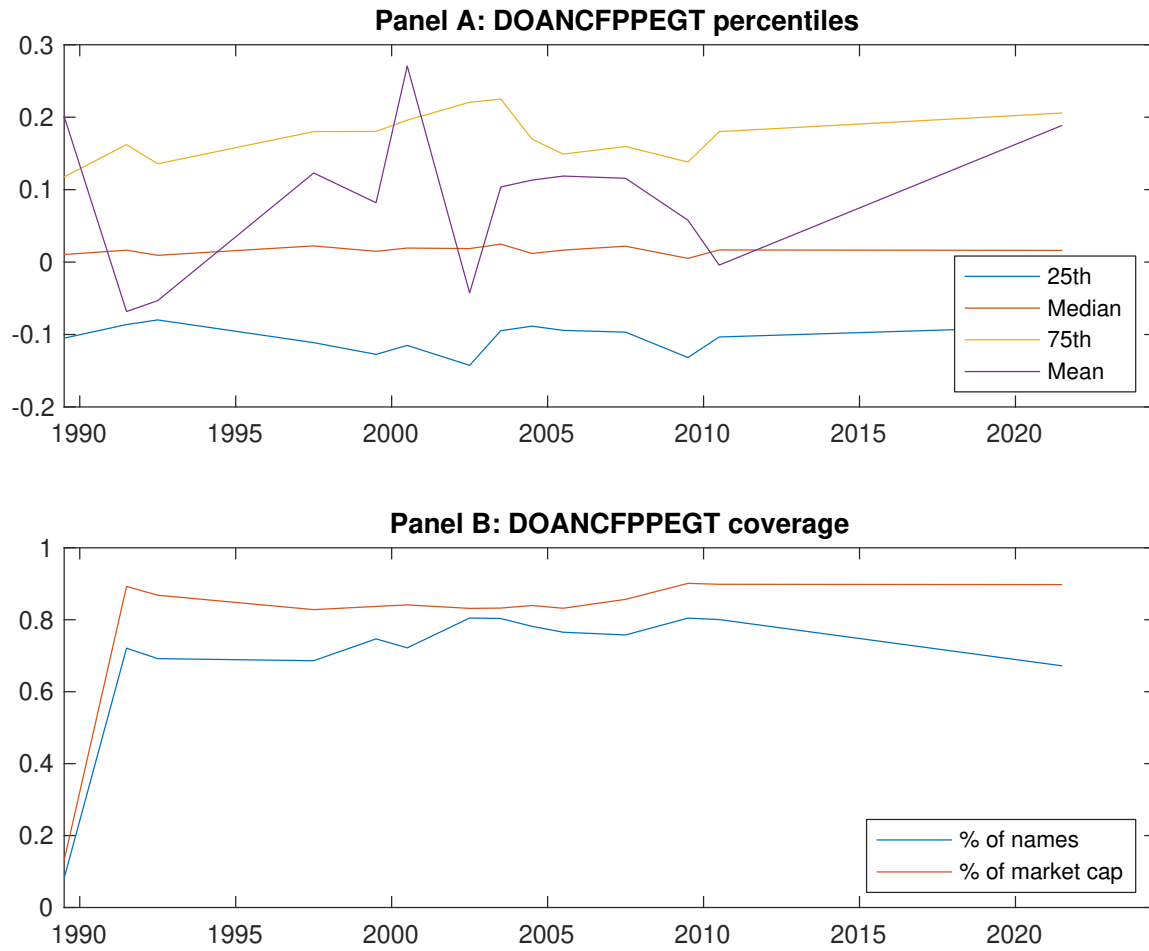
The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, ACI adds to our understanding of how cash management efficiency relates to expected returns and challenges existing asset pricing models to incorporate this dimension of firm behavior. For practitioners, the signal’s robust performance after transaction costs suggests potential real-world applicability, though implementation should carefully consider market

capacity constraints and the signal’s performance across different market conditions.

Several limitations of this study warrant acknowledgment. First, while our analysis accounts for transaction costs, we do not explicitly model market impact or capacity constraints that might affect the scalability of ACI-based strategies. Second, the sample period and market coverage may influence the generalizability of our results, particularly in emerging markets or during extreme market conditions. Third, the economic mechanisms driving the ACI premium remain partially unexplained, suggesting the need for theoretical development.

Future research should pursue several promising directions. Investigation into the underlying economic channels through which asset cash intensity affects returns would enhance our theoretical understanding of this anomaly. Time-varying analysis of the ACI premium could reveal whether the signal’s effectiveness depends on macroeconomic conditions or market states. International evidence would test the robustness of our findings across different regulatory environments and accounting standards. Additionally, examining the interaction between ACI and corporate governance or management quality might uncover important moderating effects. Finally, exploring whether machine learning techniques can enhance the signal’s predictive power by capturing non-linear relationships or interactions with other firm characteristics represents a natural extension of this work.

In conclusion, Asset Cash Intensity emerges as a robust and economically significant predictor of stock returns that contributes meaningfully to our understanding of cross-sectional return variation. While questions remain about its theoretical foundations and practical limits, the signal’s strong empirical performance suggests it deserves serious consideration in both academic asset pricing research and practical portfolio management.



**Figure 1:** Times series of ACI percentiles and coverage. This figure plots descriptive statistics for ACI. Panel A shows cross-sectional percentiles of ACI over the sample. Panel B plots the monthly coverage of ACI relative to the universe of CRSP stocks with available market capitalizations.

**Table 1:** Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on ACI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 198906 to 202406.

| Panel A: Excess returns and alphas on ACI-sorted portfolios                       |                  |                  |                  |                  |                  |                  |
|-----------------------------------------------------------------------------------|------------------|------------------|------------------|------------------|------------------|------------------|
|                                                                                   | (L)              | (2)              | (3)              | (4)              | (H)              | (H-L)            |
| $r^e$                                                                             | 0.63<br>[2.52]   | 0.71<br>[3.53]   | 0.63<br>[3.43]   | 0.70<br>[3.32]   | 0.97<br>[3.64]   | 0.33<br>[2.84]   |
| $\alpha_{CAPM}$                                                                   | -0.19<br>[-2.43] | 0.08<br>[0.93]   | 0.06<br>[0.72]   | 0.01<br>[0.09]   | 0.10<br>[1.16]   | 0.29<br>[2.47]   |
| $\alpha_{FF3}$                                                                    | -0.17<br>[-2.33] | 0.04<br>[0.55]   | 0.01<br>[0.14]   | -0.01<br>[-0.10] | 0.17<br>[2.56]   | 0.35<br>[3.24]   |
| $\alpha_{FF4}$                                                                    | -0.13<br>[-1.77] | 0.05<br>[0.66]   | -0.00<br>[-0.03] | -0.02<br>[-0.26] | 0.18<br>[2.65]   | 0.31<br>[2.90]   |
| $\alpha_{FF5}$                                                                    | -0.11<br>[-1.58] | -0.08<br>[-1.02] | -0.15<br>[-2.39] | -0.13<br>[-2.41] | 0.29<br>[4.52]   | 0.40<br>[4.01]   |
| $\alpha_{FF6}$                                                                    | -0.08<br>[-1.08] | -0.06<br>[-0.78] | -0.15<br>[-2.34] | -0.14<br>[-2.42] | 0.29<br>[4.40]   | 0.36<br>[3.59]   |
| Panel B: Fama and French (2018) 6-factor model loadings for ACI-sorted portfolios |                  |                  |                  |                  |                  |                  |
| $\beta_{MKT}$                                                                     | 1.07<br>[60.42]  | 0.92<br>[48.04]  | 0.88<br>[55.12]  | 1.01<br>[72.75]  | 1.08<br>[66.95]  | 0.01<br>[0.37]   |
| $\beta_{SMB}$                                                                     | 0.10<br>[3.93]   | 0.04<br>[1.29]   | -0.07<br>[-2.94] | -0.05<br>[-2.52] | 0.03<br>[1.40]   | -0.07<br>[-1.88] |
| $\beta_{HML}$                                                                     | -0.10<br>[-3.28] | 0.02<br>[0.69]   | 0.05<br>[1.77]   | -0.08<br>[-3.17] | -0.19<br>[-6.93] | -0.09<br>[-2.14] |
| $\beta_{RMW}$                                                                     | -0.20<br>[-6.34] | 0.13<br>[3.74]   | 0.22<br>[7.81]   | 0.23<br>[9.08]   | -0.08<br>[-2.73] | 0.12<br>[2.72]   |
| $\beta_{CMA}$                                                                     | 0.19<br>[4.16]   | 0.30<br>[6.20]   | 0.27<br>[6.75]   | 0.11<br>[3.11]   | -0.37<br>[-9.13] | -0.56<br>[-8.82] |
| $\beta_{UMD}$                                                                     | -0.06<br>[-3.76] | -0.03<br>[-1.67] | -0.00<br>[-0.16] | 0.00<br>[0.23]   | 0.01<br>[0.53]   | 0.07<br>[3.00]   |
| Panel C: Average number of firms ( $n$ ) and market capitalization ( $me$ )       |                  |                  |                  |                  |                  |                  |
| $n$                                                                               | 1006             | 481              | 432              | 533              | 972              |                  |
| $me$ (\$10 <sup>6</sup> )                                                         | 2146             | 2458             | 3229             | 3567             | 3941             |                  |



**Table 2:** Robustness to sorting methodology & trading costs

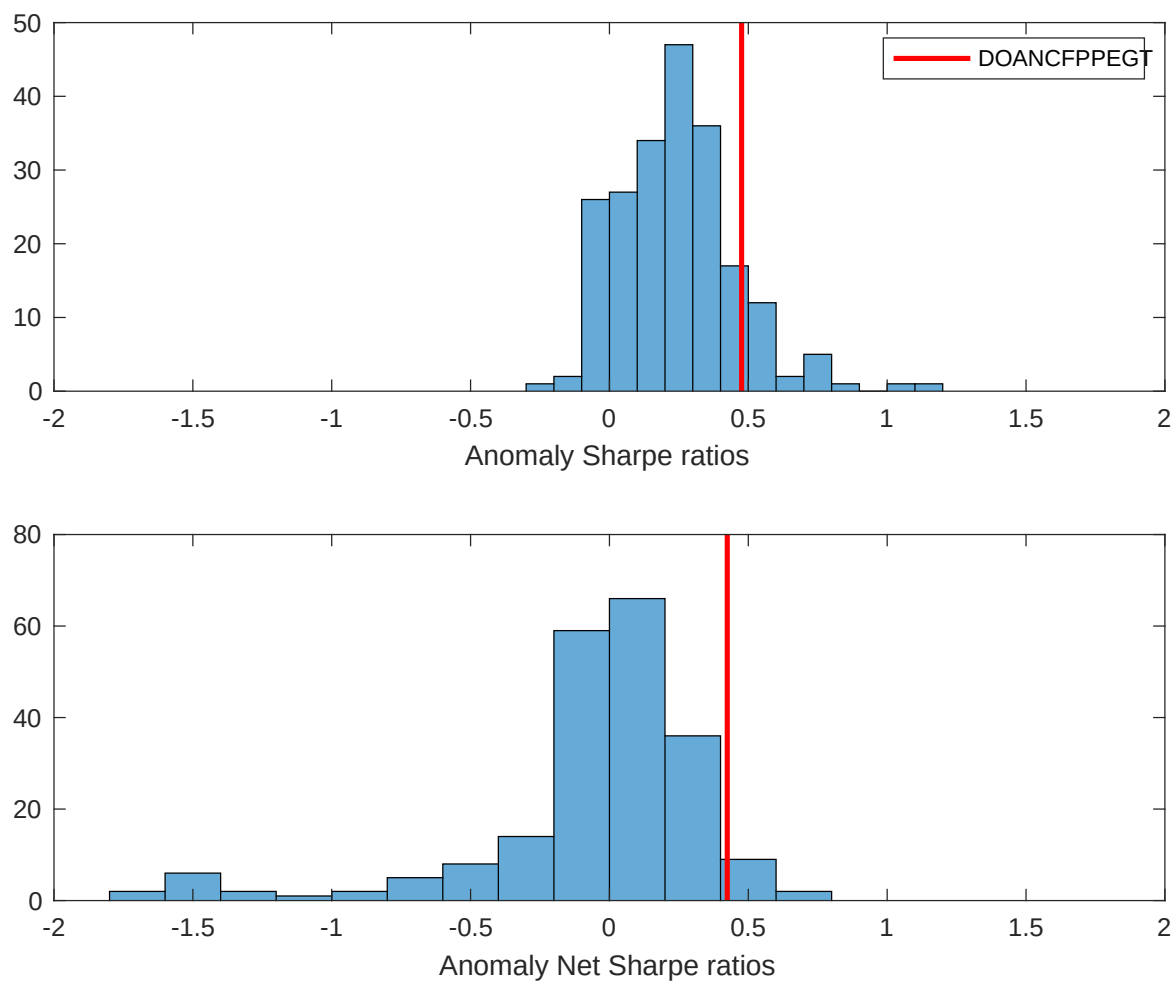
This table evaluates the robustness of the choices made in the ACI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 198906 to 202406.

| Panel A: Gross Returns and Alphas                                        |        |         |                    |                          |                         |                         |                         |                         |
|--------------------------------------------------------------------------|--------|---------|--------------------|--------------------------|-------------------------|-------------------------|-------------------------|-------------------------|
| Portfolios                                                               | Breaks | Weights | $r^e$              | $\alpha_{\text{CAPM}}$   | $\alpha_{\text{FF3}}$   | $\alpha_{\text{FF4}}$   | $\alpha_{\text{FF5}}$   | $\alpha_{\text{FF6}}$   |
| Quintile                                                                 | NYSE   | VW      | 0.33<br>[2.84]     | 0.29<br>[2.47]           | 0.35<br>[3.24]          | 0.31<br>[2.90]          | 0.40<br>[4.01]          | 0.36<br>[3.59]          |
| Quintile                                                                 | NYSE   | EW      | 0.14<br>[1.97]     | 0.17<br>[2.40]           | 0.17<br>[2.45]          | 0.10<br>[1.53]          | 0.09<br>[1.31]          | 0.04<br>[0.54]          |
| Quintile                                                                 | Name   | VW      | 0.33<br>[2.37]     | 0.30<br>[2.09]           | 0.34<br>[2.60]          | 0.30<br>[2.28]          | 0.35<br>[2.71]          | 0.30<br>[2.37]          |
| Quintile                                                                 | Cap    | VW      | 0.27<br>[1.97]     | 0.16<br>[1.15]           | 0.23<br>[1.98]          | 0.24<br>[2.01]          | 0.41<br>[3.82]          | 0.40<br>[3.63]          |
| Decile                                                                   | NYSE   | VW      | 0.35<br>[2.39]     | 0.30<br>[2.01]           | 0.34<br>[2.50]          | 0.31<br>[2.22]          | 0.37<br>[2.76]          | 0.33<br>[2.45]          |
| Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas |        |         |                    |                          |                         |                         |                         |                         |
| Portfolios                                                               | Breaks | Weights | $r_{\text{net}}^e$ | $\alpha_{\text{CAPM}}^*$ | $\alpha_{\text{FF3}}^*$ | $\alpha_{\text{FF4}}^*$ | $\alpha_{\text{FF5}}^*$ | $\alpha_{\text{FF6}}^*$ |
| Quintile                                                                 | NYSE   | VW      | 0.30<br>[2.53]     | 0.25<br>[2.10]           | 0.29<br>[2.76]          | 0.27<br>[2.59]          | 0.33<br>[3.36]          | 0.31<br>[3.13]          |
| Quintile                                                                 | NYSE   | EW      | -0.11<br>[-1.23]   |                          |                         |                         |                         |                         |
| Quintile                                                                 | Name   | VW      | 0.28<br>[2.04]     | 0.24<br>[1.70]           | 0.27<br>[2.12]          | 0.25<br>[1.96]          | 0.27<br>[2.15]          | 0.25<br>[1.97]          |
| Quintile                                                                 | Cap    | VW      | 0.24<br>[1.73]     | 0.10<br>[0.74]           | 0.16<br>[1.39]          | 0.17<br>[1.44]          | 0.32<br>[2.93]          | 0.31<br>[2.84]          |
| Decile                                                                   | NYSE   | VW      | 0.30<br>[2.08]     | 0.24<br>[1.62]           | 0.28<br>[2.04]          | 0.26<br>[1.91]          | 0.30<br>[2.23]          | 0.28<br>[2.06]          |

**Table 3:** Conditional sort on size and ACI

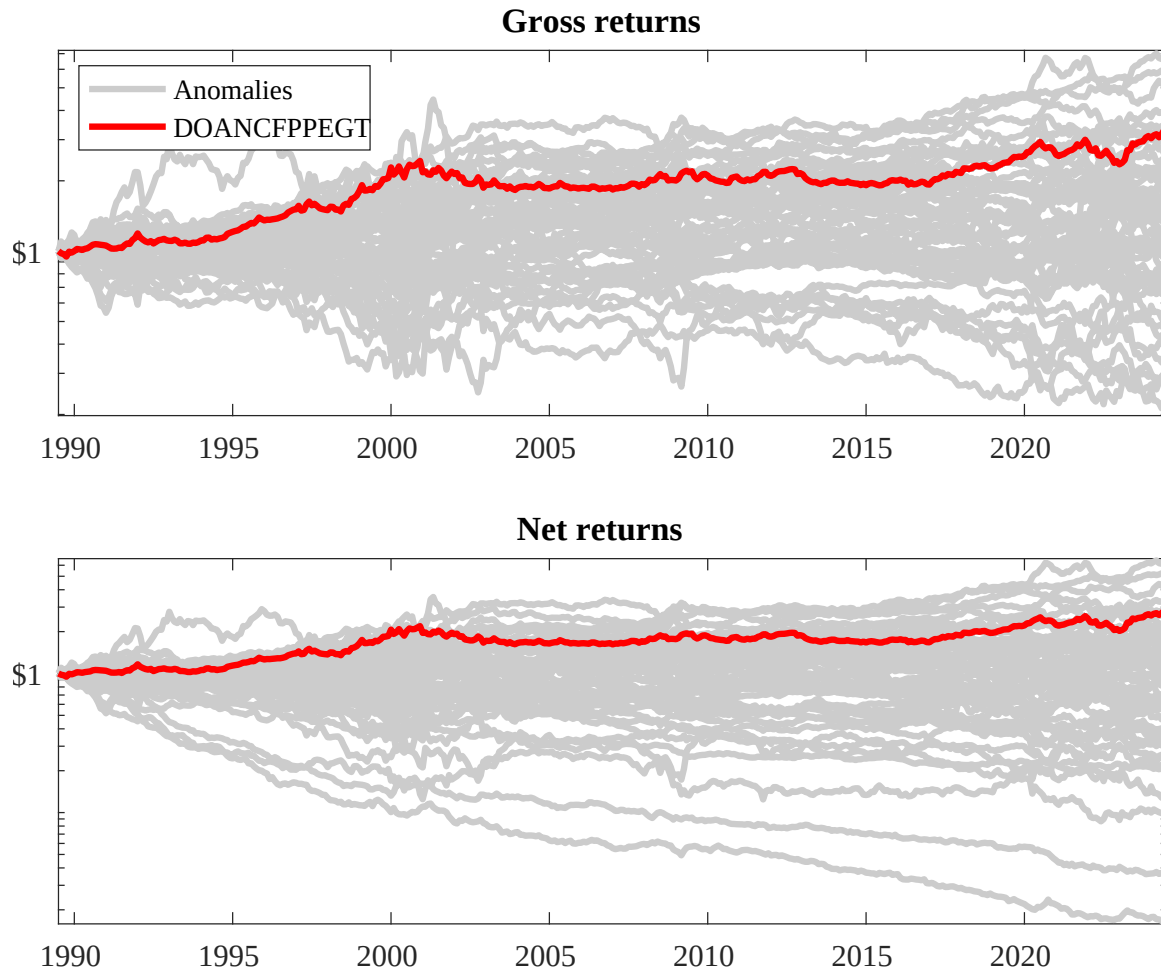
This table presents results for conditional double sorts on size and ACI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on ACI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high ACI and short stocks with low ACI. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 198906 to 202406.

| Panel A: portfolio average returns and time-series regression results |               |                |                |                |                |                |                                                    |                 |                |                |                  |                  |
|-----------------------------------------------------------------------|---------------|----------------|----------------|----------------|----------------|----------------|----------------------------------------------------|-----------------|----------------|----------------|------------------|------------------|
| Size quintiles                                                        | ACI Quintiles |                |                |                |                | ACI Strategies |                                                    |                 |                |                |                  |                  |
|                                                                       |               | (L)            | (2)            | (3)            | (4)            | (H)            | $r^e$                                              | $\alpha_{CAPM}$ | $\alpha_{FF3}$ | $\alpha_{FF4}$ | $\alpha_{FF5}$   | $\alpha_{FF6}$   |
|                                                                       | (1)           | 0.64<br>[1.62] | 0.95<br>[2.61] | 0.97<br>[3.21] | 0.88<br>[2.84] | 0.74<br>[2.06] | 0.11<br>[0.75]                                     | 0.14<br>[0.96]  | 0.09<br>[0.66] | 0.07<br>[0.53] | -0.08<br>[-0.57] | -0.09<br>[-0.64] |
|                                                                       | (2)           | 0.61<br>[1.71] | 0.78<br>[2.64] | 0.91<br>[3.32] | 0.88<br>[3.05] | 0.78<br>[2.48] | 0.17<br>[1.33]                                     | 0.25<br>[1.93]  | 0.22<br>[1.74] | 0.18<br>[1.38] | 0.08<br>[0.61]   | 0.04<br>[0.32]   |
|                                                                       | (3)           | 0.69<br>[2.26] | 0.79<br>[3.09] | 0.77<br>[3.13] | 0.90<br>[3.28] | 0.94<br>[2.98] | 0.25<br>[2.00]                                     | 0.25<br>[2.01]  | 0.27<br>[2.20] | 0.20<br>[1.61] | 0.22<br>[1.78]   | 0.16<br>[1.29]   |
|                                                                       | (4)           | 0.81<br>[2.84] | 0.76<br>[3.23] | 0.78<br>[3.43] | 0.77<br>[3.15] | 0.99<br>[3.31] | 0.18<br>[1.55]                                     | 0.16<br>[1.38]  | 0.21<br>[1.87] | 0.16<br>[1.38] | 0.26<br>[2.25]   | 0.21<br>[1.82]   |
|                                                                       | (5)           | 0.65<br>[2.78] | 0.67<br>[3.69] | 0.59<br>[3.04] | 0.73<br>[3.36] | 0.97<br>[3.46] | 0.32<br>[2.12]                                     | 0.20<br>[1.34]  | 0.27<br>[1.98] | 0.25<br>[1.79] | 0.40<br>[3.09]   | 0.37<br>[2.78]   |
| Panel B: Portfolio average number of firms and market capitalization  |               |                |                |                |                |                |                                                    |                 |                |                |                  |                  |
| Size quintiles                                                        | ACI Quintiles |                |                |                |                | ACI Quintiles  |                                                    |                 |                |                |                  |                  |
|                                                                       |               | Average $n$    |                |                |                |                | Average market capitalization (\$10 <sup>6</sup> ) |                 |                |                |                  |                  |
|                                                                       |               | (L)            | (2)            | (3)            | (4)            | (H)            | (L)                                                | (2)             | (3)            | (4)            | (H)              |                  |
|                                                                       | (1)           | 363            | 364            | 367            | 367            | 364            | 38                                                 | 42              | 46             | 45             | 40               |                  |
|                                                                       | (2)           | 112            | 112            | 113            | 113            | 112            | 76                                                 | 79              | 81             | 82             | 79               |                  |
|                                                                       | (3)           | 78             | 79             | 79             | 78             | 78             | 138                                                | 142             | 142            | 141            | 140              |                  |
|                                                                       | (4)           | 68             | 68             | 68             | 68             | 67             | 308                                                | 314             | 318            | 319            | 316              |                  |
| (5)                                                                   | 61            | 62             | 61             | 62             | 62             | 2151           | 2252                                               | 2585            | 3043           | 2425           |                  |                  |



**Figure 2:** Distribution of Sharpe ratios.

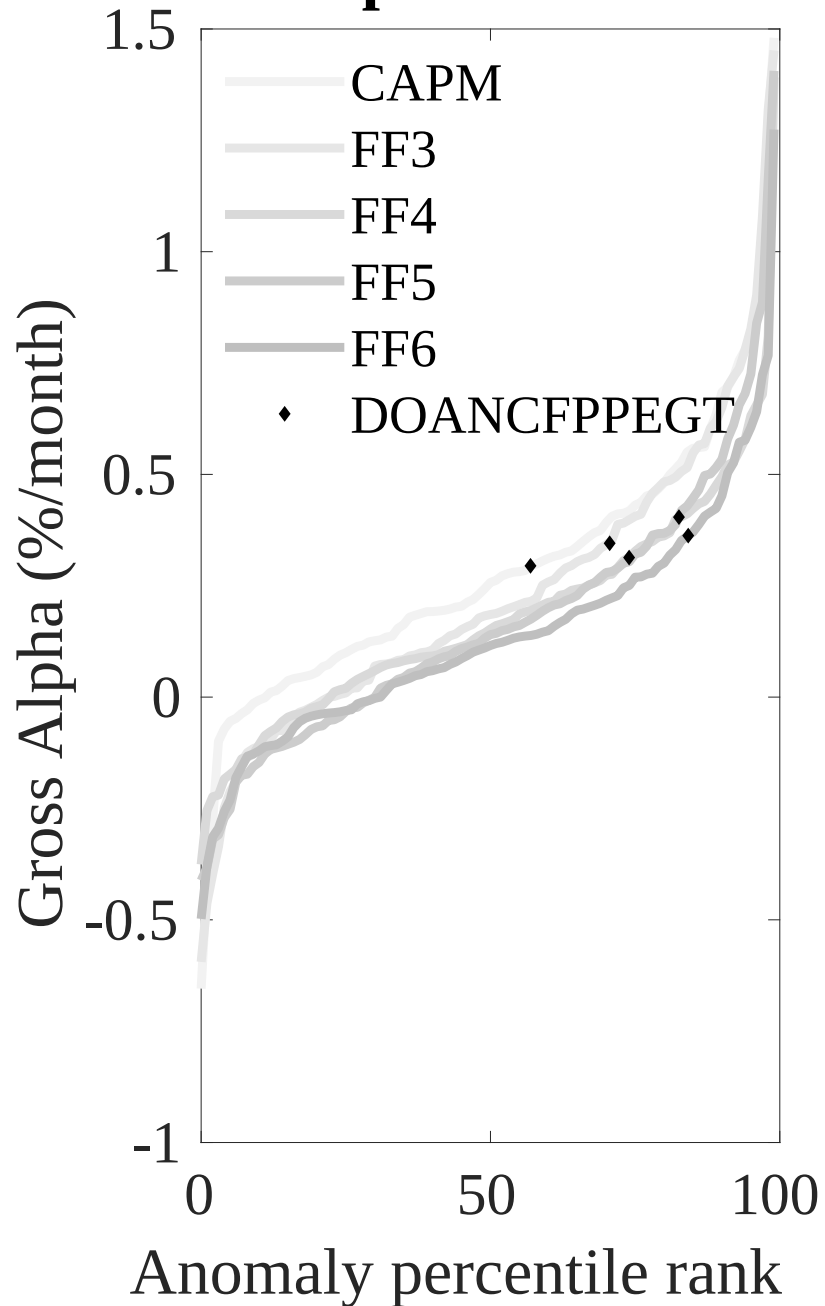
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the ACI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.



**Figure 3:** Dollar invested.

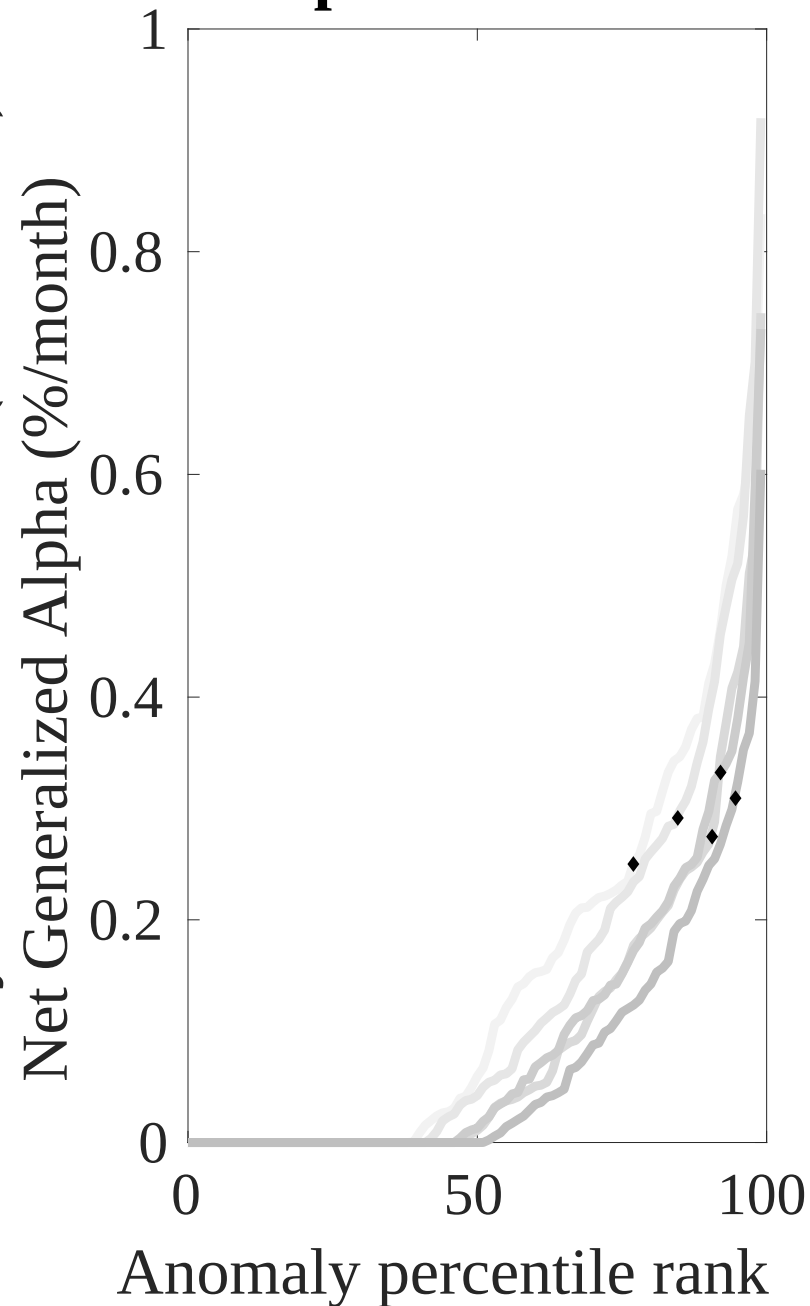
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the ACI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

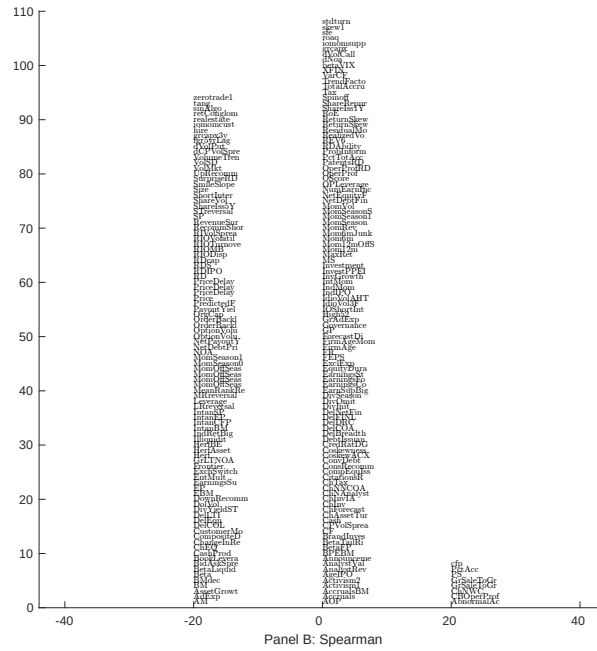
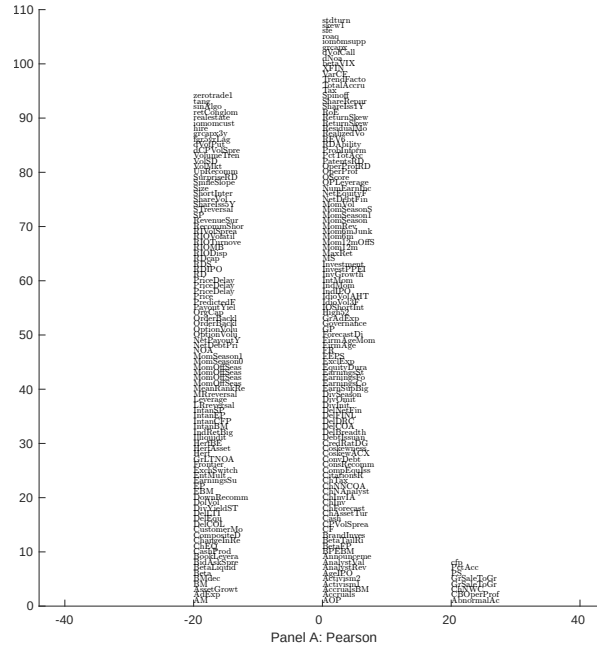
## Gross Alpha Percentiles



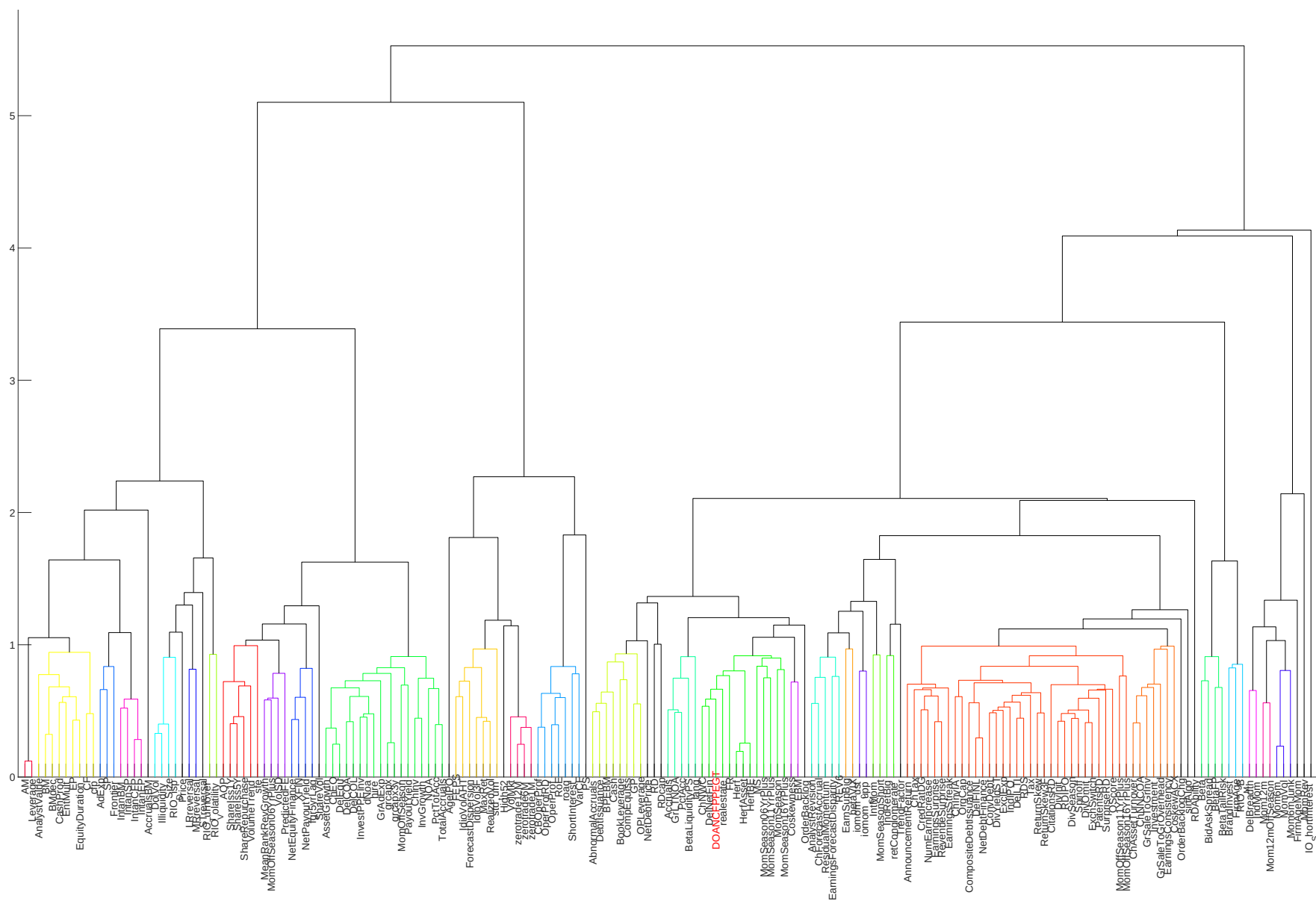
## Net Alpha Percentiles

Novy-Marx and Velikov (RFS, 2016)



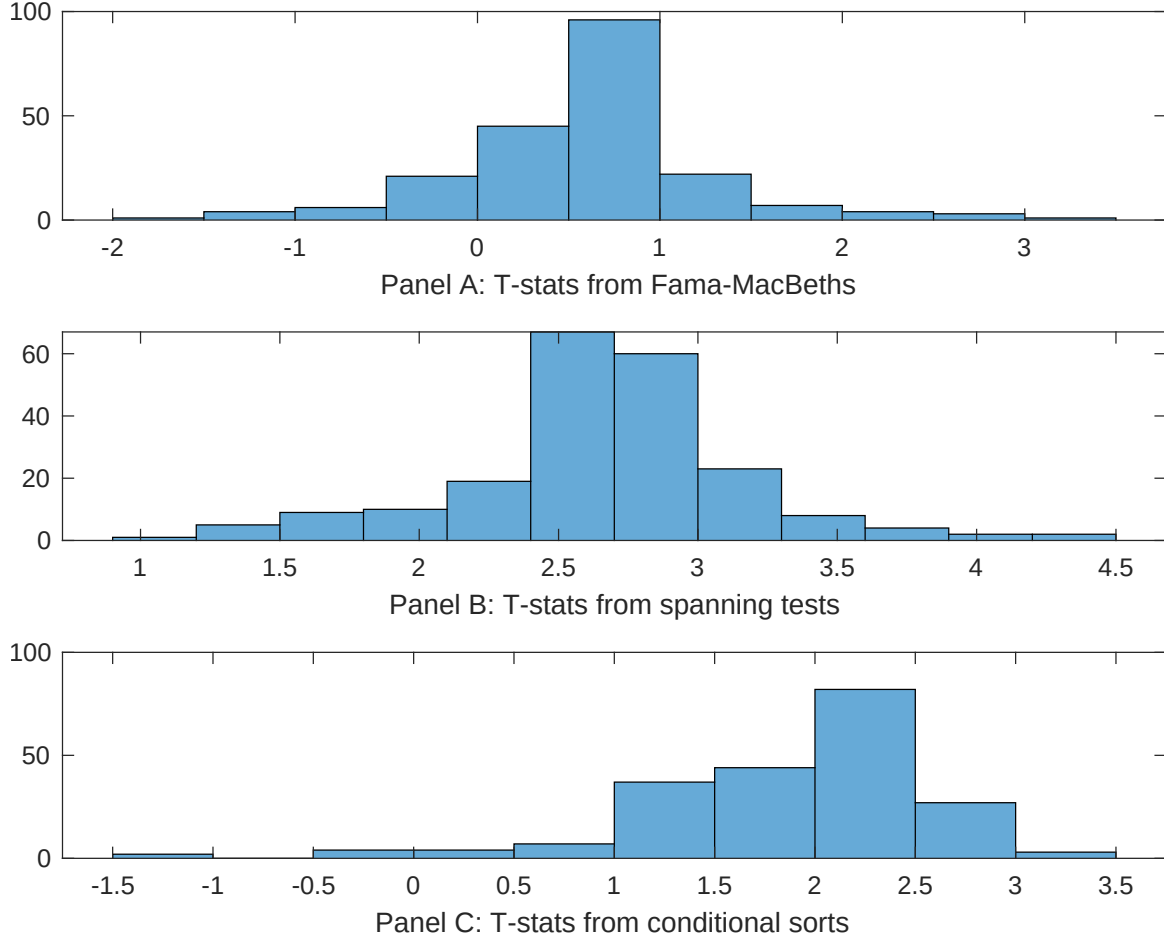


**Figure 5:** Distribution of correlations. This figure plots a name histogram of correlations of 210 filtered anomaly signals with ACI. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.



**Figure 6:** Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.



**Figure 7:** Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of ACI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on  $\beta_{ACI}$  from Fama-MacBeth regressions of the form  $r_{i,t} = \alpha + \beta_{ACI}ACI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$ , where  $X$  stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on  $\alpha$  from spanning tests of the form:  $r_{ACI,t} = \alpha + \beta r_{X,t} + \epsilon_t$ , where  $r_{X,t}$  stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on ACI. Stocks are finally grouped into five ACI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ACI trading strategies conditioned on each of the 210 filtered anomalies.



**Table 4:** Fama-MacBeths controlling for most closely related anomalies

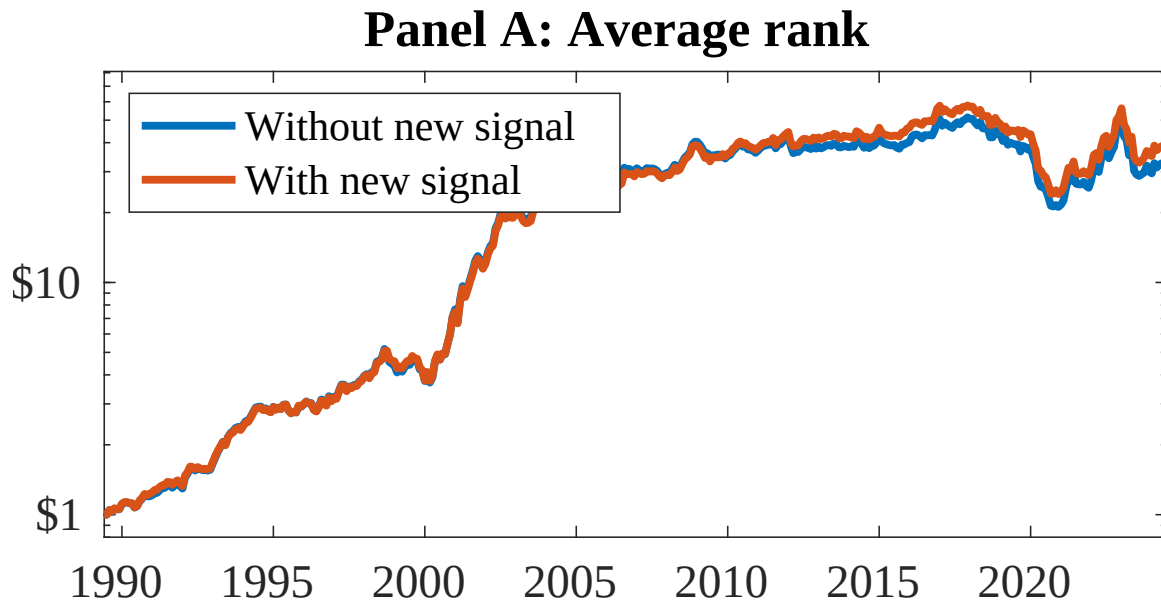
This table presents Fama-MacBeth results of returns on ACI. and the six most closely related anomalies. The regressions take the following form:  $r_{i,t} = \alpha + \beta_{ACI}ACI_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$ . The six most closely related anomalies,  $X$ , are Change in equity to assets, Growth in book equity, Change in current operating liabilities, Abnormal Accruals, Operating Cash flows to price, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the  $R^2$  from the spanning tests in Figure 7, Panel B. The sample period is 198906 to 202406.

|                 |                |                |                |                |                |                |                  |
|-----------------|----------------|----------------|----------------|----------------|----------------|----------------|------------------|
| Intercept       | 0.12<br>[3.97] | 0.17<br>[5.65] | 0.12<br>[3.94] | 0.11<br>[3.84] | 0.11<br>[3.64] | 0.13<br>[4.28] | 0.12<br>[4.53]   |
| ACI             | 0.26<br>[1.26] | 0.11<br>[0.51] | 0.16<br>[0.78] | 0.41<br>[0.17] | 0.46<br>[0.00] | 0.96<br>[0.47] | 0.35<br>[0.17]   |
| Anomaly 1       | 0.13<br>[4.51] |                |                |                |                |                | 0.92<br>[2.13]   |
| Anomaly 2       |                | 0.48<br>[6.27] |                |                |                |                | -0.25<br>[-0.30] |
| Anomaly 3       |                |                | 0.21<br>[4.41] |                |                |                | 0.75<br>[1.40]   |
| Anomaly 4       |                |                |                | 0.58<br>[0.19] |                |                | -0.92<br>[-0.27] |
| Anomaly 5       |                |                |                |                | 0.95<br>[3.11] |                | 0.78<br>[2.42]   |
| Anomaly 6       |                |                |                |                |                | 0.87<br>[7.58] | 0.52<br>[5.03]   |
| # months        | 420            | 420            | 420            | 420            | 415            | 420            | 415              |
| $\bar{R}^2(\%)$ | 0              | 0              | 0              | 0              | 1              | 0              | 0                |

**Table 5:** Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the ACI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form:  $r_t^{ACI} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$ , where  $X_k$  indicates each of the six most-closely related anomalies and  $f_j$  indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies,  $X$ , are Change in equity to assets, Growth in book equity, Change in current operating liabilities, Abnormal Accruals, Operating Cash flows to price, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the  $R^2$  from the spanning tests in Figure 7, Panel B. The sample period is 198906 to 202406.

|                 |                   |                   |                   |                   |                   |                   |                   |
|-----------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| Intercept       | 0.37<br>[3.75]    | 0.36<br>[3.64]    | 0.25<br>[2.64]    | 0.27<br>[2.79]    | 0.35<br>[3.43]    | 0.36<br>[3.72]    | 0.19<br>[2.00]    |
| Anomaly 1       | -18.31<br>[-3.62] |                   |                   |                   |                   |                   | -11.83<br>[-1.38] |
| Anomaly 2       |                   | -7.96<br>[-1.48]  |                   |                   |                   |                   | 14.74<br>[1.82]   |
| Anomaly 3       |                   |                   | -34.74<br>[-7.99] |                   |                   |                   | -26.85<br>[-5.60] |
| Anomaly 4       |                   |                   |                   | 27.06<br>[6.08]   |                   |                   | 19.42<br>[4.29]   |
| Anomaly 5       |                   |                   |                   |                   | 0.95<br>[0.24]    |                   | 1.84<br>[0.50]    |
| Anomaly 6       |                   |                   |                   |                   |                   | -22.95<br>[-4.46] | -9.36<br>[-1.40]  |
| mkt             | 0.04<br>[0.02]    | 0.07<br>[0.03]    | 1.09<br>[0.47]    | -2.51<br>[-1.03]  | 0.55<br>[0.22]    | 0.12<br>[0.05]    | -1.01<br>[-0.44]  |
| smb             | -8.06<br>[-2.30]  | -7.37<br>[-2.07]  | -10.90<br>[-3.27] | -9.57<br>[-2.80]  | -7.87<br>[-2.17]  | -6.79<br>[-1.95]  | -12.35<br>[-3.69] |
| hml             | -8.60<br>[-2.00]  | -9.78<br>[-2.25]  | 6.56<br>[1.44]    | -6.83<br>[-1.63]  | -10.61<br>[-2.07] | -8.60<br>[-2.02]  | 4.91<br>[0.95]    |
| rmw             | 11.93<br>[2.69]   | 13.10<br>[2.92]   | 13.67<br>[3.27]   | 16.35<br>[3.76]   | 12.09<br>[2.59]   | 12.39<br>[2.82]   | 13.20<br>[3.06]   |
| cma             | -36.71<br>[-4.58] | -47.68<br>[-5.93] | -36.92<br>[-5.87] | -49.02<br>[-8.02] | -55.00<br>[-8.55] | -27.87<br>[-3.22] | -26.93<br>[-3.10] |
| umd             | 6.40<br>[2.96]    | 6.51<br>[2.96]    | 4.55<br>[2.22]    | 5.12<br>[2.42]    | 6.84<br>[2.42]    | 5.74<br>[2.67]    | 4.43<br>[1.69]    |
| # months        | 420               | 420               | 420               | 420               | 416               | 420               | 416               |
| $\bar{R}^2(\%)$ | 38                | 36                | 45                | 41                | 36                | 39                | 47                |



**Figure 8:** Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 158 anomalies. The red solid lines indicate combination trading strategies that utilize the 158 anomalies as well as ACI. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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